

# Xuanyu Li

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## EDUCATION

**Cornell University**

*Ph. D, Department of Mathematics*

**Ithaca, NY, US**

*2023.9-Present*

**University of Science and Technology of China(USTC)**

*Undergraduate, School of Mathematical Science*

**Hefei, Anhui, China**

*2019.8-2023.7*

## RESEARCH INTERESTS

Minimal surfaces and harmonic maps, especially their singularity structures and existence theorem.

## PREPRINTS AND PUBLICATIONS

Li X. Optimal regularity of stable harmonic maps to spheres. arXiv preprint arXiv:2608.20272 (2026).

Gaia, F. and Li, X. Existence of constant mean curvature surfaces with controlled topology in 3-manifolds. arXiv preprint arXiv:2602.16635 (2026).

Li, X. Existence of expanding harmonic map flows to hemispheres. arXiv preprint arXiv:2602.08932 (2026).

Li, X. Energy identity for Ginzburg-Landau approximation of harmonic maps. arXiv preprint arXiv:2503.23675 (2025).

Li, X. Minkowski content estimates for generic area minimizing hypersurfaces. Calc. Var. 63, 176 (2024).

## AWARDS AND GRANTS

**Simons Dissertation Fellowship in Mathematics**

*Simons Foundation, 2026*

**ICM 2026 Travel Grants for U.S.-Based Graduate Students**

*American Mathematical Society (AMS), 2026*

**Robert John Böttig Graduate Prize**

*Cornell, 2025*

**Guo MoRuo Scholarship (highest distinction at USTC)**

*USTC, 2023*

## Service

I have reviewed a paper for Journal of Differential Geometry.

## TALKS

**Uniqueness and non-uniqueness results of harmonic map flow**

PDE seminar, USTC

*06/2026*

Union College Mathematics Conference 2026 (20 mins)

*09/2026*

**Existence of CMC surfaces with bounded genus**

Geometric Analysis Seminar, CUHK

*07/2026*

Geometric PDES and minimal surfaces workshop, Shanghai Center for Mathematical Science

*07/2026*

Geometry-Topology seminar, UPenn

*03/2026*

**Eneyg identity for Ginzburg-Landau approximation of harmonic maps**

Geometric Analysis seminar, USTC

*06/2025*

Geometric Analysis seminar, Zhejiang University

*06/2025*

## TEACHING

**Math4130 Honors Introduction to Analysis I (Grader)**

*2026 Fall*

**Math6160 Partial differential equations (Grader)**

*2026 Spring*

**Math3040 Prove it! (Recitation TA)**

*2026 Spring*

**Math6520 Differential manifolds (Recitation TA)**

*2025 Fall*

**Math4180 Complex analysis (Grader)**

*2025 Spring*

**Math2930 Differential equations for engineers (Recitation TA)**

*2024 Fall*

**Math2930 Differential equations for engineers (Recitation TA)**

*2024 Spring*